Exp: 2

Learning / Assignment of IP

25/07/26

AIM:

To Understand the IP addressing scheme & manually assign static IP addresses to computers on a network.

PROCEDURE:

1. In WINDOWS:

   Open network & sharing centre → change Adapter setting → local area connection → properties IPv4 → set IP Manually.

2. Enter IP : 192.168.1.10
   Subnet mask : 255.255.255.0
   Default gateway: 192.168.1.1
3. Enter DNS : 8.8.8.8, click OK
4. In LINUX:

   Use command 'sudo ip addr add 192.168.1.10/24 dev eth0'.
5. Set gateway : 'sudo ip route add default via 192.168.1.1'.
6. Verify : 'ipconfig' or 'ip a'
7. Ping the default to confirm network connectivity.
8. Ping another computer on the same subnet to verify communication.

COMMANDS :

```
sudo ip addr add 192.168.1.10/24 dev eth0
sudo ip route add default via 192.168.1.1

ip a show eth0
ping 192.168.1.1
nmcli con mod 'Wired connection1' ipv4.
                    addresses 192.168.1.10/24
nmcli con mod 'Wired connection 1' ipv4.
                    gateway 192.168.1.1
nmcli con mod 'Wired connection 1' ipv4.
                    method manual
nmcli con up 'Wired connection 1'
```



RESULT :

Static IP addresses were successfully assigned to computers. Connectivity was verified using ping commands b/w hosts.